

```

import heapq
def display_board(state):
    """Display the 8-puzzle state in 3x3 matrix format."""
    for i in range(0, 9, 3):
        print(state[i], state[i+1], state[i+2])
    print()
def misplacedtiles(state, goalstate):
    """Heuristic: number of misplaced tiles (excluding blank)."""
    count = 0
    for i in range(9):
        if state[i] != 0 and state[i] != goal_state[i]:
            count += 1
    return count
def getnextstates(state):
    """Generate all possible next states."""
    index = state.index(0)
    r, c = index // 3, index % 3
    moves = [(0, 1), (0, -1), (1, 0), (-1, 0)] # Right, Left, Down, Up
    next_states = []
    for dr, dc in moves:
        nr, nc = r + dr, c + dc
        if 0 <= nr < 3 and 0 <= nc < 3:
            new_index = nr * 3 + nc
            new_state = list(state)
            newstate[index], newstate[new_index] =
            newstate[newindex], new_state[index]
            nextstates.append(tuple(newstate))
    return next_states
def bestfirstsearch(initialstate, goalstate):
    """Solve 8-puzzle using Best First Search."""
    open_list = []
    visited = set()
    heapq.heappush(
        open_list,
        (misplacedtiles(initialstate, goalstate), initialstate, 0) )
    while open_list:
        h, currentstate, steps = heapq.heappop(openlist)
        if currentstate == goalstate:
            return steps
        visited.add(current_state)
        for nextstate in getnextstates(currentstate):
            if next_state not in visited:
                heapq.heappush(
                    open_list,

```

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(misplacedtiles(nextstate, goal_state),  
next_state, steps + 1)  
)  
return None
```

----- Main Execution -----

```
initial_state = (1, 2, 3,  
4, 5, 6,  
0, 7, 8)  
goal_state = (1, 2, 3,  
4, 5, 6,  
7, 8, 0)  
print("Initial State:")  
displayboard(initialstate)  
print("Goal State:")  
displayboard(goalstate)  
steps = bestfirstsearch(initialstate, goalstate)  
if steps is not None:  
print("Final Number of Steps Required:", steps) else:  
print("No solution found")
```